

Grammaticality de-idealized

Brett Reynolds 

Humber Polytechnic & University of Toronto

brett.reynolds@humber.ca

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Abstract

Why is *I have 25 years* easy to understand but ungrammatical as a way of stating one's age in English, while *Colorless green ideas sleep furiously* is grammatical despite its odd meaning? I argue that grammaticality research often runs together four questions. Does the language provide an analysis for the expression? Do the grammatical contributions of its parts fit together? Does it mark the contrasts required in this context? Is the construction established in this speech community and situation? Processing difficulty and implausibility are separate: they can lower acceptability without changing grammatical status. The Operator-Value Model of Grammaticality (OVMG) represents grammaticality as a structured profile of these conditions rather than a single scalar. It also separates the population profile, a speaker's or analyst's evidence about it, and the resulting feelings of anomaly and confidence. These distinctions predict that rarely encountered constructions should respond more to exposure than forms repeatedly displaced by an established competitor; that absence is evidence of non-licensing only where the missing form was a plausible choice; that wrong values in paradigms that strongly organize what an utterance adds to the conversation should prompt a different repair locus from value-intact form errors; and that evidence for a moribund contrast should weaken before its categorical treatment changes. A conditional support-and-change module models how use, competition, forgetting, adoption, and retention can alter population licensing. Its simulations establish model-internal possibilities under specified assumptions, not mechanisms in actual language communities. The proposal replaces the ideal speaker's single grammaticality fact with distinct, testable claims about structure, convention, evidence, speaker experience, and change.

Keywords: grammaticality; projectibility; acceptability; preemption; entrenchment; conventionalization

I INTRODUCTION

Every competent speaker of English knows that **Can the have running* is impossible, but the source of this certainty proves remarkably elusive. What does it mean to say a sentence is ungrammatical? Consider these examples:

- (1) a. **Can the have running?*
- b. *Colorless green ideas sleep furiously.* (Chomsky, 1957)
- c. **I've finished it yesterday.*
- d. *?I saw Joan, a friend of whose was visiting.*
- e. *The bread the baker the apprentice helped made is delicious.*
- f. A: *How old are you?* B: **I have 25 years.*
- g. **Which did you buy car?*

These examples don't fail in the same way. (1a) resists a covering analysis; (1c) combines incompatible grammatical instructions; and (1f) is interpretable but lacks licensing for age predication in English. Independent relative *whose* has unsettled status, whereas centre embedding is structurally licensed but hard to process. Left-branch extraction is different again: a short, interpretable pattern that English speakers reject with striking confidence.

Recent Danish evidence makes the need for separation concrete. Christensen and Nyvad (2024) varied structural complexity, lexical and construction frequency, and acceptability in one design. Complexity predicted acceptability better than construction frequency, while zero or near-zero frequency covered sharply different acceptability levels. That result doesn't validate OVMG or settle licensing, but it shows why structure, processing, frequency, and acceptability can't be treated as rival values on one scale.

The de-idealization in the title is specific. Early generative grammar located grammaticality in the knowledge of an ideal speaker–listener in a homogeneous community (Chomsky, 1965, p. 3). That idealization bought tractability by treating variation, uncertainty, and change as external to the competence fact. Following Goodman (1955), I instead ask first what the category GRAMMATICAL lets us predict—its projectibility—and then what profile supports those predictions and what might stabilize it (Boyd, 1999; Khalidi, 2013, 2018; Millikan, 1999; Reynolds, 2026a; Slater, 2015). Naming a mechanism doesn't by itself settle either question.

The Operator-Value Model of Grammaticality (OVMG) develops that answer. It first identifies paradigms by closed alternatives, a recurring domain, a semantic frame, and obligatory value selection. It then grades how strongly their values organize what an utterance adds to the conversation. The high-profile end is the operator stratum; tone, particles, word order, and inflection can all realize its values (Reynolds, 2026b). This stratum is narrower than grammaticality as a whole. The framework rests on three premises: profile, projection, and support.

1. Profile. A grammaticality claim is shorthand for a profile: a bundle of answers to several separable questions about a candidate in a situation. Is there any available analysis for it? Do its grammatically encoded contrasts agree with one another? Are the contrasts required in that situation supplied? Is the pattern a conventional resource for the relevant community, dialect, or register? And, separately, does it merely feel bad because it's implausible or hard to process? Section 3 develops

these as coverage, compatibility, saturation, licensing, and subjective read-out. The examples in (1) don't instantiate one kind of "ungrammaticality"; they occupy different profiles.

2. Projection. The profile earns its standing by what it lets us predict, not by the mechanisms that hold it together: which rejections should soften under repeated exposure and which resist (1d) vs. (1g), how faithfully forms transmit, and how they change over time.
3. Support, tested separately. The dynamics module specifies candidate sufficient conditions under which usage could stabilize a population licensing state; conversational repair is a candidate *controller* – a process that detects a deviation and helps correct it – testable by its intervention signature (§4.4); and the subjective read-out of ungrammaticality is separate from grammatical status: it comprises gradient anomaly and confidence signals.

The individual ingredients have substantial precedents in generative, constructional, usage-based, sociolinguistic, and interactional work. The contribution claimed here is their integration into one architecture that keeps population conditions, evidence, subjective response, and causal support apart and makes each distinction answer to held-out predictions and failure conditions.

Section 2 states the problem. Sections 3–4 give the reader-facing state and dynamics theories before the diagnostic contrasts apply them in detail. Section 6 then states the predictions and positions the account against neighbouring frameworks. The accompanying supplement contains the notation, derivations, and simulations. Its Lean component checks only structural consequences of the definitions—including separations among conditioning, opportunity, operator identity, value, exponent choice, and licensing—not the population dynamics or empirical classifications.

2 WHY A SINGLE GRAMMATICALITY VARIABLE IS INSUFFICIENT

Three familiar problems show why a single grammaticality variable is insufficient. They concern whether an analysis exists, whether its grammatical instructions agree, and whether community convention supports it. The last divides into two questions in the model: whether the niche requires a contrast to be marked, and whether the community licenses the construction used to mark it.

2.1 STRUCTURAL VIABILITY

Formal production systems make membership in a set of well-formed strings categorical (Chomsky, 1957; Post, 1943). That captures cases such as (1a), which resist a viable structural analysis. It doesn't by itself distinguish them from analyzable but degraded strings. Assigning every such contrast to performance risks the circularity that Schütze (2016, p. 71) identifies: troublesome judgments can be set aside precisely because they trouble the proposed competence grammar.

2.2 COMPATIBILITY OF GRAMMATICAL INSTRUCTIONS

Form and meaning can also come apart in two directions. (1b) has intact grammatical instructions despite an implausible construal, whereas the intended message in (1c) is recoverable despite a clash between the present perfect and *yesterday*. Generative semantics and Construction Grammar established that grammatical constructions make semantic contributions (Goldberg, 1995; Lakoff, 1971; McCawley, 1968); the remaining problem is to separate clashes among grammatical instructions from lexical implausibility and processing cost.

2.3 SITUATIONAL LICENSING

Finally, structurally viable and interpretable expressions can remain outside a community's repertoire. Sociolinguistic and usage-based work locates such patterns in community convention, entrenchment,

and statistical preemption (Bybee, 2006; Goldberg, 2011; Labov, 1972). The age-predication use in (1f) is rejected in English but licensed by cognate-looking frames in French and Spanish. Structural coverage, compatibility, community convention, and felt acceptability require different diagnostics and evidence streams. The next section states those distinctions before applying them.

3 FROM AN UTTERANCE TO A GRAMMATICALITY PROFILE

The opening examples show why one acceptability scale can't do all the work. This section states the model in ordinary language; the formal supplement gives the specification, derivations, simulations, and notation. The main idea is simple: a grammaticality claim compresses several answers about a form–value pairing in a particular community and situation. The answers can come apart, and their pattern matters more than any single score.

The model separates a STATE THEORY from a DYNAMICS THEORY. The state theory asks what has to be true for a candidate to count as grammatical now. The dynamics theory asks how learning, choice, competition, repair, and forgetting could make that state persist or change. This separation blocks a common shortcut: a story about how a pattern arose doesn't by itself establish the pattern's present status, and a stable present pattern doesn't identify the mechanism that produced it.

3.1 FIX THE SITUATION BEFORE DIAGNOSING THE FORM

Every diagnosis is relative to a CONDITIONING STATE: the communicative situation as the participants construe it, together with the community, variety, register, or norm centre—the standard to which they orient. This state can include the activity, medium, footing, attributed speaker identity, and relevant discourse community. It isn't assumed to be obvious or externally given. Interlocutors can construe the same encounter differently, and analysts can draw the boundary too broadly or too narrowly.

For that reason, the conditioning state has to be specified independently of the judgment the model is meant to explain. A study can't label every accepting speaker as belonging to one variety and every rejecting speaker as belonging to another after seeing the responses. It needs prior evidence from community membership, usage, interactional cues, self-identification, or a registered coding protocol. If an independently motivated partition turns an apparently divided population into internally stable groups, the original conditioning state was underspecified. If no such partition does, the disagreement is a genuine population fact.

3.2 OPERATOR VALUES AND PUBLIC UPDATE

The term VALUE has the Saussurean sense of *valeur*: a linguistic unit is identified by what it contrasts with, what it patterns with, and which interpretations it makes available (Saussure, 1916). It doesn't mean a feature assignment or decision-theoretic value. OVMG treats grammatical knowledge as conditioned form–value relations at multiple levels of grain.

Some grammatical contrasts do more than distinguish expressions. They help determine how an utterance enters the conversational record: whether it counts as an assertion or question, who did what to whom, when it holds, what follows from it, and which referents or commitments later contributions can take up. OVMG calls these OPERATOR VALUES and their contribution a PUBLIC-UPDATE ROLE.

Operator analysis proceeds in two stages. A paradigm has a closed set of alternatives, a recurring domain, a language-specific semantic frame, and obligatory value selection whenever that domain ap-

plies. Identification is categorical for this comparison. An identified paradigm then receives a graded operator profile: how strongly its values configure public update across ordinary occasions of use. The operator stratum is the high-profile end. Illustrative candidates include clause type, polarity, temporal anchoring, agreement and role allocation, scope, and reference tracking. The profile applies across ordinary uses of a paradigm, not to every token. Tone, particles, word order, inflection, and suprasegmental material can realize its values (Reynolds, 2026b).

The formal core needs a discrete inventory when it checks whether grammatical instructions unify. It uses independently identified paradigms and a task-specific cut on the operator profile for clear high-profile cases. The response model retains the magnitude of public-update contribution. That cut serves the discrete interface; it doesn't turn the graded profile into a sharp natural boundary.

The operator stratum is narrower than grammaticality. A collocation or head-specific complement construction can be categorically licensed or excluded without belonging to an operator paradigm. A moribund paradigm can retain a high operator profile while its licensing evidence becomes less settled. Opportunity affects that evidence, not paradigm status or the operator profile. Attribution fails if, after paradigm structure, opportunity, and competitor strength are matched, a low-profile paradigm produces the same update confusion and update-oriented repair as a high-profile one. The descriptive profile may remain useful, but it adds no causal reach beyond constructional licensing and entrenchment.

3.3 FOUR CONSTITUTIVE QUESTIONS

OVMG treats constructions in the broad Construction Grammar sense: learned, typed associations among form, structure, content, context, and conventional constraints, at any grain from an inflectional cell to a clause pattern or an intonational contour (Fillmore et al., 1988; Goldberg, 1995; Sag, 2012). A candidate utterance can normally be assembled in more than one way from such resources. The model asks whether *some* analysis of the intended form and value passes four tests.

1. Coverage: can the language build it? There has to be at least one complete analysis of the form with the intended value. Word salad may fail here. A merely novel sentence need not: productive constructions can cover a token that has never previously occurred.
2. Compatibility: do its operator instructions fit together? The contributions that tell interlocutors how to update the conversation have to be jointly satisfiable. In **I've finished it yesterday*, the present perfect and the deictic past-time adverb provide incompatible temporal instructions. More exposure to the same analysis can't average that conflict away.
3. Saturation: are the contrasts required in this niche supplied? Here a niche is a recurring communicative job, defined more fully in §4.1. A community can come to treat overt marking of a distinction as obligatory. On this account, obligatoriness is the historical result of situations in which an unmarked alternative repeatedly loses to an overtly marked one. A moribund distinction can later cease to be obligatory.
4. Licensing: are the constructions used in the analysis established resources here? A form can be analyzable, interpretable, and internally compatible without being part of the relevant repertoire. English **I have 25 years* illustrates a head-and-complement convention of this kind. The intended age meaning is recoverable, but English licenses a different frame. No operator has to be mis-set for a construction to fail this test.

These tests are ordered for diagnosis but aren't four degrees on one scale. Coverage and compat-

ibility are properties of analyses in a specified inventory. Saturation records a path-dependent community convention. Licensing is a population pattern over constructional resources. A candidate is grammatical when at least one analysis passes all four; alternative failed analyses don't contaminate a successful one. This scope matters for ambiguity, coercion, and productive novelty.

The distinction between grammatical value and exponent choice cuts across the four tests. **Goed* expresses the intended past value but selects an unlicensed exponent—the form chosen to realize that value. Turkish suffix-harmony errors have the same value-intact shape: the plural or past value remains identifiable, while the chosen allomorph isn't licensed in that phonological environment (Supplementary File 2). Whether the containing system meets the paradigm criteria and receives a high operator profile requires separate evidence. Conversely, **depend of* is a head-specific complement convention, not an operator error. Neither a finite inventory nor categorical rejection establishes paradigmhood or a high operator profile.

3.4 POPULATION STATUS AND A SPEAKER'S EVIDENCE

The four questions concern a population profile, but no speaker or analyst sees that profile directly. Each observes a history of tokens, alternatives, omissions, repairs, and contextual cues and forms an estimate from it. OVMG keeps three things distinct: the community's current licensing pattern, an individual speaker's evidence-based estimate of that pattern, and an analyst's estimate from a possibly different sample.

This distinction adds an important second dimension to the familiar gradient mean. Two forms can receive the same middling rating for quite different reasons. One speaker may have little evidence either way; another community may be sharply divided between stable users and stable non-users. The first is epistemic uncertainty and should be sensitive to framing and new exposure. The second is population heterogeneity and should instead appear as stable between-speaker disagreement. OVMG calls the amount and consistency of evidence behind an estimate its EVIDENCE CONCENTRATION: high concentration means that the estimate is well settled for or against licensing, not merely that its average is high. Mean ratings alone erase these differences.

The model also distinguishes uncertainty about the constructional inventory from uncertainty about licensing. If analysts don't know whether a productive analysis exists, that's an uncertainty about coverage. Once a viable analysis is fixed, sparse evidence about whether a community uses its nodes is a licensing question. Treating every unfamiliar token as absent from the inventory would wrongly predict that first-heard but readily productive sentences begin as ungrammatical.

3.5 WHAT THE PROFILE IS EXPECTED TO PREDICT

The profile earns its theoretical role only if observing part of it licenses expectations about cases not already used to define it. In plain terms, the analysis should tell us more than which label to attach to yesterday's rating. It should help predict how a form will be repaired, learned, transmitted, softened by exposure, or destabilized over time.

The central projection contrasts weak evidence with strong contrary evidence. A construction rare because its niche seldom occurs should remain uncertain and comparatively movable under informative exposure. A form absent despite many genuine opportunities, and repeatedly displaced by an established competitor, should produce confident and framing-resistant rejection. A processing-heavy but licensed sentence should improve with practice without a corresponding change in production or repair evidence. A moribund contrast should first lose evidential support and only later

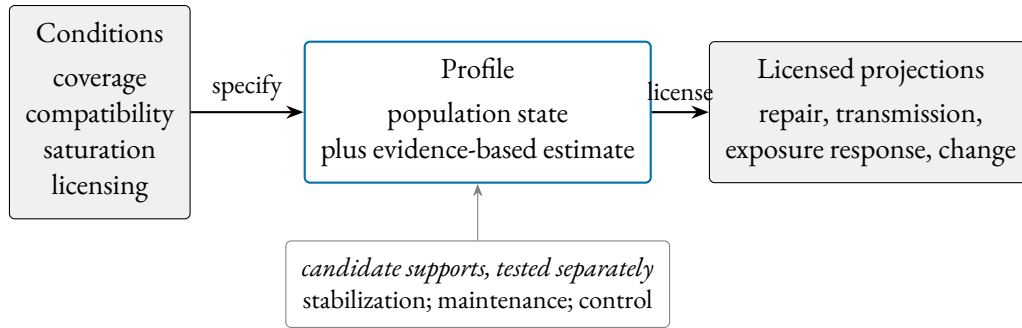


Figure 1: The projectibility-first order of explanation. The constitutive conditions specify a four-condition population profile. Evidence about that profile, including how settled the evidence is and how much speakers differ, licenses the projections on the right. Stabilization, maintenance, and control are separate world-side hypotheses (below), each requiring its own evidence.

lose its categorical treatment. These targets go beyond the tests used to assign the initial profile and can fail independently.

Those failures have to change the analysis. The situation, niche, and target projection should be fixed before the outcome is inspected. If a profile then fails to predict held-out repair, exposure, transmission, or change, the projection has to be narrowed, split, or abandoned. It can't be rescued by redrawing the speech-community boundary after the responses are known. If independently measured profile components neither dissociate nor improve held-out prediction over a single latent score, the multipart profile itself hasn't earned its role.

Stabilization, maintenance, and control are distinct explanatory claims. A stable profile might be the residue of history, the result of ongoing transmission, or the effect of an active corrective process. Evidence for one doesn't establish the next. Repair remains a candidate controller until intervention shows that a deviation is registered and that the response helps restore the threatened higher-level relation.

3.6 STATUS AND THE FEELING OF STATUS

Speakers don't inspect a population state. They experience an utterance and respond to it. OVMG represents that response with two ordinary-language dimensions: how anomalous the token feels and how confident the speaker is in that reaction. Low expected licensing can contribute to anomaly, but so can processing load, lexical implausibility, garden paths, prescriptive pressure, and misidentification of the speaker or situation.

Confidence supplies information that anomaly alone misses. A preempted gap may elicit a confident "that's impossible". A winnerless cell may elicit an equally low rating but the phenomenology "I don't know how to say it". An independent-relative *whose* may feel unfamiliar while remaining movable under framing. Centre embedding can feel very bad even when the speaker accepts the construction once its parse is made clear. These aren't verbal glosses added after the fact: the model predicts different patterns of test-retest stability, confidence, exposure response, production, and repair.

A community or institution may impose a practical acceptance boundary on this graded evidence. That boundary belongs to a decision layer: changing the costs of false acceptance and false rejection

should change how a hearer responds without changing the underlying licensing profile. A sharp switch in overt response would test the decision process, not prove that grammatical knowledge itself contains a numerical cutoff.

3.7 MISALIGNMENT AND THE SCOPE OF A SUCCESSFUL ANALYSIS

Hearer and speaker can disagree about which situation is in force. A form licensed in one dialect, genre, age group, or bilingual practice may be heard through another community's expectations. The resulting anomaly can reflect a real mismatch between two stable systems rather than uncertainty within either one. Attribution matters too: a hearer who treats a token as belonging to another community, a learner error, noise, or personal overload need not respond as if it threatened the local convention.

Within a fixed situation, the existence of one successful analysis is enough. This prevents the model from rejecting ordinary ambiguity merely because one available parse fails. But the value has to be fixed independently enough to make the claim vulnerable. If analysts are free to invent a new value whenever an analysis would otherwise fail, coverage becomes vacuous. The intended value should be constrained by context, paraphrase, comprehension, and the speaker's subsequent commitments.

3.8 MEASUREMENT REQUIRES CONVERGING CHANNELS

Status isn't directly observed, so no single observation defines it. The empirical programme combines four channels:

- Production shows which available alternatives speakers choose in independently specified niches. A low corpus rate can reflect unavailability, low preference, or avoidance, so production alone can't identify licensing. Separate acceptability and gap-filling tasks make the distinction visible: a placement can receive low isolated acceptability but still be strongly preferred to the offered alternative (Nogueira Sánchez, 2025).
- Repair shows how interlocutors respond to a putative violation. Understanding repair, update-oriented repair, overt correction, and prescriptive sanction have to be coded separately.
- Judgment and confidence measure the subjective read-out rather than status directly. Repetition, framing, and test–retest designs show how stable that read-out is.
- Situation cues support the analyst's and hearer's identification of the relevant community and activity. They can't be inferred solely from the target rating.

A strong test requires at least one non-judgment indicator of licensing, an independent measure of whether changing an operator value changes the public update, and a niche definition fixed without using the target's acceptability. The formal supplement states the corresponding observation model and its identification assumptions. The agreement corpus probe reported there illustrates the discipline: a rate per opportunity is evidence about choice, and becomes evidence about licensing only after the alternatives and avoidance option have been independently normed.

3.9 THE OPENING EXAMPLES REVISITED

The profile now gives the opening asterisks different explanations. The results are summarized in Table 1; the labels describe the decisive condition rather than mutually exclusive sentence types.

4 HOW LICENSING PATTERNS MAY CHANGE

The state theory says what a present profile consists of. It doesn't explain how the profile came to be. The conditional support-and-change module supplies candidate routes, with two constraints on

Case	Decisive diagnosis	Expected profile
<i>*Can the have running</i>	no covering analysis	confident rejection across construals
<i>Colorless green ideas sleep</i>	licensed but implausible	grammatical status with content oddness
<i>furiously</i>	content	
<i>*I've finished it yesterday</i>	incompatible temporal	confident rejection resistant to exposure
	instructions	of the parts
<i>*I have 25 years</i>	concentrated	recoverable meaning but confident,
	non-licensing of the	competitor-backed rejection
	English age frame	
<i>'friend of whose</i>	sparse opportunity and	low confidence and comparatively strong
	weak evidence	framing or exposure effects
<i>*Which did you buy car?</i>	putative	confident rejection only if the
	high-opportunity	opportunity and competitor
	preemption	assumptions are independently met
<i>pobedit' 'win', first-person</i>	winnerless cell (no	low rating with ineffability rather than
<i>singular</i>	conventional winner)	confident correction
<i>centre embedding</i>	implementation overload	practice can improve the experience
		without changing licensing

Table 1: Reader-facing OVMG profiles. Each diagnosis makes a different prediction beyond the judgment used to introduce it.

the explanation. First, learning about a community isn't the same event as changing that community. Second, absence is informative only when the missing candidate had a genuine opportunity to occur.

4.1 OPPORTUNITY, CANDIDATES, AND THE OUTSIDE OPTION

A NICHE is a recurring communicative job, such as expressing a recipient, locating an event in time, or marking an information source. Its candidate set contains the realizations speakers could plausibly choose for that job at that time. It also contains an outside option: paraphrase, clarification, silence, or avoidance. The outside option matters because failure to produce a target doesn't always mean that a competitor defeated it.

Production reflects both licensing and choice. A licensed form may be rare because speakers prefer another licensed form. An unlicensed form may be absent for the same surface reason. Conversely, a form may be scarcely attested simply because the relevant niche is scarcely encountered. The analyst has to establish the opportunity set, candidate preferences, and avoidance rate without using the target's grammaticality judgment as the coding criterion.

The evidence stream distinguishes three events. A direct error provides evidence about the target itself. An omission provides evidence only to the extent that the target would plausibly have been chosen had it been available and that the niche was correctly identified. Repair provides a further observation whose meaning depends on its type and on the relation between the error and the public update. These events enter one shared model but aren't interchangeable counts.

4.2 LEARNING AND POPULATION CHANGE

An individual updates an estimate when evidence arrives. The population changes when that evidence alters later inclusion, production, learning, or transmission. Updating an evidence-based estim-

ate can make beliefs more precise; adoption and abandonment require a further population process. The formal dynamics specify that process, whose costs have to be motivated independently.

Depending on the adoption and retention settings, weakly supported constructions may disappear while strongly supported ones persist, producing separated population patterns. Other settings yield a single interior pattern from the same learning rule. Categorical-looking regimes depend on empirically challengeable assumptions beyond exposure.

Cross-sectional comparison can measure the conventional fixation that this process is intended to explain. In storyboard experiments across six Oceanic languages, Franjeh et al. (2025) find a cline from variable possessive classification to more than 90% matching use in North Ambrym. That profile operationalizes fixed assignment; it doesn't by itself establish the historical transition or the adoption–retention mechanism proposed here.

Familiar motivations for change enter through those assumptions rather than as alternative definitions of grammaticality. Semantic reanalysis can make a new form–value relation usable, as in the development of the *going to* futurate (Bybee et al., 1994). Social indexing can likewise change adoption and retention. Processing economy, analogy, and iconicity can likewise alter opportunities and choices. Optimality Theory has productively represented competing pressures as ranked constraints (Prince & Smolensky, 2004); OVMG instead asks how independently motivated changes in opportunity, preference, adoption, or retention alter a community's licensing profile.

4.3 WHAT THE SIMULATIONS ESTABLISH

The simulations are existence and coherence checks. Each speaker has a licensing estimate and a binary repertoire state. Each round, choices among the target, competitors, and an outside option update the estimate. The evidence-only control fixes repertoire states; coupled runs let the estimate affect later adoption and retention. In the reported sweep, linear adoption stayed interior, while a threshold and steep smooth response produced separated patterns; finite populations recovered the separation. A separate search found two stable states divided by an unstable one and distinct slow-sweep loss and recovery boundaries.

A predeclared early-warning check failed: variance and lagged correlation didn't rise reliably, so no early-warning prediction follows. These results don't establish bounded-memory long-run behaviour, residence times, network or cohort effects, or a shared-latent observation model. The supplement records the parameters, outputs, and open proof and identification obligations.

4.4 REPAIR AS A CANDIDATE MAINTAINER OR CONTROLLER

Repair can feed back into learning, but its targets differ. A repair may secure understanding without endorsing the form, replace an exponent while preserving the intended operator value, or correct an operator mis-setting. Overt correction can also reflect prescriptive or identity policing rather than a threat to the update.

Operator attribution predicts a selective response profile; paradigm identification and profiling alone don't guarantee one. Once paradigm structure, opportunity, competitor strength, and ordinary comprehensibility are controlled, mis-setting an operator value in a high-profile paradigm should disproportionately elicit update-oriented repair. A categorically unlicensed but non-operator construction may instead be corrected as a conventional form choice. The prediction concerns the repair target. Non-operators can still be categorically policed.

Calling repair a *controller* requires an intervention showing that the system registers a deviation, changes lower-level behaviour, and thereby restores the threatened population relation. Showing only that removing repair weakens transmission would establish maintenance, not control.

4.5 FOUR PREDICTED TRAJECTORIES

The dynamics yield four qualitatively different trajectories that would be collapsed by a single frequency measure.

OBLIGATORIFICATION. An overt contrast becomes obligatory when unmarked realizations repeatedly lose in well-attested niches and the pattern becomes conventional. The claim is historical and niche-specific: neither present uncertainty nor simple absence diagnoses obligatoriness, and sustained recovery has to precede its loss. A contrast that became obligatory without repeated losses in a well-attested niche, or ceased to be obligatory without prior evidential decline, would weaken the obligatorification mechanism.

WINNERLESS CELLS. A dense niche can lack a conventional winner. Paraphrase or avoidance then starves all candidates of evidence without, by itself, lowering support for one of them. Low licensing with low confidence also requires independently motivated weak starting support or diffuse negative evidence. The result is ineffability rather than a competitor-backed ban. Russian defective paradigms supply the clean model; English **amn't* is probably mixed.

DESTABILIZATION. A moribund contrast can lose evidential concentration before its average status changes much. As shared evidence disappears, individual histories can drift toward starting expectations. Early between-speaker divergence requires heterogeneous networks, cohorts, priors, or exposure histories: concentration loss is the direct prediction; dispersion is conditional.

ACTUATION AND APPARENT CATEGORICAL EXCLUSIONS. Changes in opportunity, competitor strength, analogy, repair, or social ascription can move a construction between long-run regimes, but identifying a historical trigger requires independent diachronic evidence. Persistent preemption in a dense niche can also concentrate rejection despite an available compatible analysis. This diagnosis requires independent evidence that the niche was rich, the target was a live candidate, and a competitor repeatedly won.

The dynamics succeed only if independently measured opportunities, alternatives, repair types, evidence histories, and adoption costs predict the distinct trajectories. The formal supplement supplies the quantitative claims and reproducible simulation contract.

5 DIAGNOSTIC CONTRASTS

The model's questions now let us compare cases without forcing them onto one severity scale. The labels below name common routes through the architecture, not additional primitives. Each case also supplies evidence or a falsifier that the short diagnoses in Table 1 leave implicit.

5.1 CONDITIONING AND COMMUNITY CONVENTION

Form–value relations occur at every level of grammatical organization, from morphemes to abstract clause patterns (Goldberg, 1995). Individual forms can participate in several such relations: English past-tense morphology usually locates an event in the past, but it can also mark social distance (*Could you help me?*) or low likelihood (*If I went tomorrow...*) (Kilgariff, 2004). A diagnosis concerns a relation in a specified situation, not a form with one context-free meaning.

Communicative situations overlap and shift with activity, audience, durable social position, and the norm centre participants orient to (Wiese, 2023, p. 3). Multiple modals such as *I might could help you with that* are productive for some American English speakers and rejected by others (Morin & Grieve, 2024). Spanish–English bilingual practice likewise licenses some recurrent mixtures while excluding others (Toribio, 2001). These sources of variation still have to be fixed independently by the procedure in §3.1.

Convergent acceptance is evidence about a licensed repertoire. Status has to be specified independently enough to identify contested innovations, stable dialect differences despite mutual intelligibility, and performance errors that pass unnoticed. The form, value, situation, and comparison class can't be inferred from the response alone.

5.2 SATURATION OF OBLIGATORY CONTRASTS

Languages differ in which distinctions their constructions require speakers to mark. Standard English normally requires progressive aspect for an event presented as ongoing (*She is studying right now*). Standard French can describe the same situation with the simple present (*Elle étudie maintenant*) because progressive marking isn't obligatory there. Evidential distinctions provide a parallel cross-linguistic contrast: some languages grammaticalize an obligatory evidential paradigm, whereas English usually leaves source marking to optional lexical or modal resources. OVMG treats these obligations as path-dependent conventions whose establishment and possible loss are addressed by the obligatorification trajectory in §4.5.

5.3 OPERATOR-VALUE COMPATIBILITY

Compatibility asks whether the operator contributions of a proposed analysis can jointly hold. The individual pieces may be familiar and the intended message recoverable even when their grammatical instructions clash.

Consider the temporal incompatibility in:

- (2) * *I've finished it yesterday.*

For the Standard English variety fixed for this illustration, the present perfect construction contributes a reference interval anchored to present relevance, while *yesterday* contributes a deictically past interval. Unlike lexically incongruous combinations like *colorless green ideas*, which remain grammatical because the clash doesn't mis-set an operator value, this example shows failed unification between grammatically encoded temporal contributions.

Sometimes compatibility failures involve conflicting information-packaging requirements:

- (3) * *Who did the lifeguard who saved _ work in New Jersey?* (Cuneo & Goldberg, 2023)

Cuneo and Goldberg (2023) analyse the same participant as simultaneously focused through fronting *who* and backgrounded by the relative-clause construction. OVMG counts this as an operator-compatibility case only if the relevant focus–background contrast is independently identified as a paradigm and assigned a high public-update profile. Otherwise it remains a typed constructional incompatibility and supplies no operator-specific prediction.

The temporal case also delimits compatibility. The condition treats operator-value conflict as failed unification among partial operator assignments: tense–aspect anchoring where tense–aspect is obligatory, argument-linking requirements where a construction licenses roles, information-packaging instructions only where they belong to an independently identified operator paradigm,

and indexical contrasts only where they've become relatively closed and update-configuring. This keeps arbitrary lexical implausibility out of grammatical incompatibility.

5.4 PROCESSING COST WITHOUT STATUS FAILURE

Processing demands can trigger feelings of ungrammaticality even when a construction passes all four status tests. Centre embedding provides the clearest contrast:

- (4) *The bread the baker the apprentice helped made is delicious.*

Dependency locality is a well-documented source of this cost: integration grows harder as dependencies span more intervening material, especially when new referents intervene (Gibson, 2000, 2026; Grodner & Gibson, 2005). Specialized language networks and domain-general systems can both contribute to the resulting experience (Fedorenko et al., 2024). Avoidance can later affect transmission, but that later route into change is separate from present status. A highly licensed construction can remain grammatical while its anomaly signal stays high.

Socio-pragmatic indexicality is distinct. Forms such as Río de la Plata *vos* and phonological cues can identify region, relationship, or stance and thereby help fix the situation and norm centre (Babel, 2025; Bertolotti, 2016; Eckert, 2012; Silverstein, 1976). An indexical contrast enters the operator stratum only where independent evidence establishes a paradigm and a strong public-update contribution.

5.5 LICENSING, OPPORTUNITY, AND PREEMPTION

Even a well-formed construction with a clear meaning can fail because the community doesn't license it in the relevant situation.

Age predication with **I have 25 years* provides the opening case. English licenses *have+years* for duration, remaining time, or experience, but ordinary age predication uses the *be* frame. The intended age is recoverable and no operator value is mis-set; the decisive failure is constructional licensing. The stronger claim that the established *be* frame supplies concentrated preemptive evidence requires independent production and repair evidence rather than the judgment alone.

- (5) **We sheared three sheeps.*

Regular plural marking has a compositionally recoverable meaning and is structurally parallel to other English plurals, but the irregular form *sheep* is entrenched across English-speaking communicative situations, making the regularized version unacceptable. Without a metaphorical or playful justification (as in *the black sheeps of the family*, where the regularized plural might signal a figurative usage), the utterance is ungrammatical. Like **goed*, the pattern is a value-intact licensing failure; the difference is grain, with a lexeme-specific plural convention displacing the productive default.

Situational licensing operates independently from the other conditions. A construction may have complete operator compatibility and minimal processing demands but still be excluded because a different form has been conventionalized for that meaning. Extreme rarity makes the distinction clear. Some constructions are so infrequent that speakers lack a shared consensus about their status.

Independent relative *whose* shows why rarity isn't automatically negative evidence. Its component resources are familiar: independent genitives are possible (*Mine was visiting*), independent interrogative *whose* is grammatical (*Whose was open?*), and dependent relative *whose* is common (*the*

student whose friend was visiting) (Hankamer & Postal, 1973). But their combination is so unusual that Hankamer and Postal (1973) deem it non-existent:

(6) ? *I saw Joan, a friend of whose was visiting.*

(adapted from Huddleston & Pullum 2002, p. 472)

The construction requires an accessible possessor, a predictable possessum, an appropriate information structure, and an ellipsis-licensing environment at once (Reynolds, 2024). Those conditions rarely converge. An exploratory COCA search found no token (Davies, 2024), but the exact interface query wasn't preserved and raw component frequency is the wrong denominator: separate occurrences of independent and relative *whose* don't create opportunities for the combined niche. The zero result supplies little negative evidence. Speakers should remain uncertain and comparatively responsive to framing or exposure, although high surprisal can add processing difficulty even for speakers who recover the analysis (Shain et al., 2020).

This contrasts with putative preempted cases like left-branch extraction, discussed next, where the relevant niche appears often and the construction is systematically untaken.

APPARENT CATEGORICAL EXCLUSIONS. Left-branch extraction is a textbook case of persistently categorical-looking rejection:

(7) * *Which did you buy [___ car]?*

Although the intended meaning is clear and processing demands aren't obviously excessive, English speakers reject the construction; Snyder (2022, p. 22) treats this pattern as a core case of "ungrammaticality".

Under OVMG, such patterns are candidate preempted gaps: situational licensing is driven toward zero by persistent preemption when the communicative job is common, an established competitor wins, and the counterfactual choice probability of the missing form is high. For English left-branch extraction, the natural competitor is *Which car did you buy?*. The corpus absence is suggestive but not decisive: the missing evidence is how often speakers would choose the discontinuous form if permissive framing made it available. Garden-path reanalysis at the bare interrogative can strengthen subjective categoricity without adding a structural constraint. The relevant dynamics are summarized in §4.5 and specified in the supplement.

Cross-linguistic comparison requires comparable niches rather than a shared label such as LEFT BRANCH (Haspelmath, 2010). Inflectional recoverability, productive discontinuity, and a useful discourse niche should favour licensing; whether case-rich, article-less languages instantiate the English construction remains open (Reynolds, 2026c). A preempted-gap diagnosis requires an independently registered high-opportunity niche, a viable covering analysis, candidate-choice norming, and persistent non-licensing in production and repair. Judgment confidence may corroborate that diagnosis, but framing and exposure response are held out for prediction.

Prediction: If left-branch extraction is a high-concentration preempted gap, framing tokens as intentional dialectal or ingroup resources should do little to move judgments; the same manipulation should matter more for low-concentration cases such as independent relative *whose* or defective cells. A substantial framing effect, evidence that speakers would rarely choose the form even if available, or failure to establish a viable covering analysis would count against the diagnosis. This tests the case assignment; the model-level test is the registered rank-order reversal in §6.1.

WINNERLESS CELLS: PARADIGM DEFECTIVENESS. Preempted gaps have a winning competitor. Defective paradigm cells don't. They leave a high-opportunity cell empty without any single form winning it. A clearer case is the Russian first-person singular non-past of *pobedit* 'win': speakers avoid the expected cell and resort to periphrasis instead. The English *amn't* gap is diagnostically useful but less clear, because broader niches include *I'm not* as a competitor, while narrower contracted-negator niches make the cell look winnerless (Broadbent, 2009; Pertsova, 2016; Pertsova & Kuznetsova, 2015; Sims, 2023; Thoms et al., 2025).

In the Russian case, the niche is common and candidate forms are readily identifiable, but no candidate accumulates enough positive evidence to settle as the form. The source can be structural uncertainty, lexical restriction, learning dynamics, or avoidance, with the mixture varying across cases (Sims, 2023). In this account, defectiveness is a third licensing profile: high opportunity with evidence divided among candidate forms, leaving no candidate stably established or confidently excluded.

Speakers should report not knowing how to say it (ineffability), rather than the confident "that's wrong" of a preempted gap. The dynamics in §4.5 explain why avoidance needn't provide confident negative evidence against any particular candidate. Defective cells should remain more movable under framing than preempted gaps.

The contrasts also explain why an intermediate rating has no unique diagnosis. It can reflect weak licensing evidence, a strong anomaly response to a licensed form, or disagreement about the conditioning state. Lexical-semantic oddity without an operator clash, as in *colorless green ideas*, likewise belongs to the subjective read-out rather than grammatical status. The profile table in §3.9 summarizes the resulting routes and their different projections.

5.6 THE SUBJECTIVE EXPERIENCE OF GRAMMATICALITY

Speakers register putative violations through a read-out that can diverge from conventional status. The anomaly channel resembles other metacognitive feelings such as the FEELING OF KNOWING (Hart, 1965); the confidence channel records how settled the speaker takes the reaction to be. The distinction also recalls Sapir's "form-feeling" and is consistent with self-monitoring in Broca's aphasia (Oomen et al., 2005; Sapir, 1921, 1927). This response can help speakers detect a violation, but it doesn't define grammaticality (Fanselow, 2021).

Status has to be inferred from converging measures, much as temperature is inferred from the behaviour of instruments. Ratings and factor analyses primarily reveal the subjective read-out; they bear on grammar only when they converge with independent indicators of licensing, compatibility, and confidence (§3.8). Raw language-model probability has the same limitation, and even minimal-pair contrasts are informative only when the intended message is controlled (Hu et al., 2026).¹

Misattribution makes the separation visible. A grammatical construction can feel wrong after a garden path:

- (8) *The old man the boats.* (Ritchie & Thompson, 1984)

The initial noun-phrase analysis of *old man* produces nonsense; on the licensed analysis, *the old* is the subject and *man* the verb. Conversely, a clear intended meaning can mask an unlicensed analysis:

¹Nefdt and Ladyman (2026) distinguish proposals that locate linguistic patterns in outputs, cognition, model-internal compression, or a plurality of models. OVMG's narrower claim is that grammaticality, for the purposes tracked here, is a projectible pattern in situation-indexed licensing profiles.

- (9) **In Michigan and Minnesota, more people found Mr. Bush's ads negative than they did Mr. Kerry's.* (Pullum, 2009)

The first clause invites a comparison between numbers of people, but the pronoun *they* in the second supplies the wrong comparison structure. Hearers often recover the intended message without registering that failure.

6 PREDICTIONS AND THEORETICAL CONSEQUENCES

The architecture earns its keep through contrasts that simpler descriptions don't predict. This section begins with those tests, then asks how the proposal relates to macro-typology and neighbouring theories. Licensing, feeling, and their possible supports retain separate evidence streams throughout.

6.1 PREDICTIONS

The model's distinctive predictions come from separating positive evidence, opportunity structure, how likely a form would have been chosen had it been licensed, how settled the licensing evidence is, and repair flow. Repair locus is the most direct test of the operator distinction; the remaining predictions concern concentration, zero-evidence learning, L2 transfer, moribundity, and change by re-licensing.

REPAIR TYPE. The repair streams (§4.4) predict which part of an utterance is targeted. Update-oriented repair should increase with niche opportunity and with the extent to which changing the contrast changes the public update, conditional on the other factors. Form correction can remain strong for a categorically excluded construction or exponent even when the intended update is preserved. Opportunity, update consequences, and licensing concentration are independently estimable from opportunity-annotated corpora, comprehension probes fixed before the repair data is inspected, and non-rating licensing streams. If matched low-profile paradigms draw the same update-oriented repair as high-profile paradigms, or if that repair tracks social indexing rather than measured update consequences, the operator attribution fails even if the descriptive profile remains useful.

SATIATION FRAMING, AS A RANK ORDER. Framing should matter most where the evidence is least concentrated. Once the relevant niche has been identified independently, confirmed preempted gaps and **sheeps* should show the least movement; entrenched but uncommon constructions should show more; and independent relative *whose* and clean winnerless cells such as *pobedit*' first-person singular should show the most. Intentional dialectal, literary, or ingroup framing should improve low-concentration forms more than error framing while leaving confirmed preempted gaps largely unmoved. Lu et al.'s (2024) meta-analysis of syntactic satiation in extraction from islands supplies a nearby benchmark for extraction phenomena, not a direct measurement of left-branch extraction. Outcomes include rating and confidence change, later repair, and production. If, under matched opportunities, low-concentration forms move less than confirmed preempted gaps, the concentration account fails.

ARTIFICIAL-LANGUAGE LEARNING. Hold positive evidence at zero while varying opportunities for an unproduced target. Competitors repeatedly occupy the high-opportunity niche; the low-opportunity niche rarely arises or would seldom favour the target. OVMG predicts rejection in the first condition and uncertainty in the second despite identical zero positive evidence. This design adapts artificial-language paradigms that manipulate simplicity and communicative structure (Culbertson & Kirby, 2016).

L2 PREEMPTION TRANSFER. Early L2 rejections should track the accumulated weight of counterfactually informative omissions in the L1, not just L2 frequency. Strong L1 preemption in a high-opportunity niche should yield early rejection of the L2 analogue; weak L1 preemption should leave rare L2 patterns unresolved. This sharpens Selinker's (1972) concept of INTERLANGUAGE: transfer should be strongest where the L1 supplies many attributable omissions. If early rejections instead track L2 frequency once L1 opportunity and competitor history are matched, the claim fails.

CONCENTRATION LOSS BEFORE CATEGORICAL DECLINE. For a contrast going moribund, the direct destabilization result (§4.5) is loss of evidence concentration as opportunities thin. Longitudinal designs should measure confidence, test–retest stability, and sensitivity to controlled exposure alongside means and production. A reliable drop in those concentration-sensitive measures while the population rate remains approximately stable would support the proposed precursor. Inter-speaker judgment dispersion is a separate, conditional target. When speakers begin with different expectations but share the same thinning evidence, the spread between them grows more slowly than the average falls. A dispersion-leading result should be predicted only when cohort replacement, unequal network opportunity loss, heterogeneous retention, or a specified judgment mapping supplies faster divergence. If registered opportunity loss doesn't reduce the concentration-sensitive measures before mean status, the proposed precursor fails; the timing of dispersion is orthogonal unless a separate heterogeneity mechanism is specified.

SOFT ALTERNATIONS AS A NON-BIMODAL CONTROL. The conditional dynamics (§4.3) predict separated licensing only where the observation process and independently motivated adoption costs support it. A both-licensed alternation should keep a unimodal licensing profile, with variation carried by selection among two variants and any outside option. The dative supplies the proposed control (§6.5) after a corpus choice model has been calibrated in its high-frequency core. Its optionality should yield confident, gradient judgments rather than the uncertainty of a winnerless cell (§4.5). If non-judgment indicators instead reveal stable licensed and unlicensed subpopulations within a preregistered niche and variety, the both-licensed diagnosis fails; stable bimodality despite a smooth, independently estimated adoption response would also defeat the dynamics prediction.

CHANGE BY RE-LICENSING. Requiring exact operator compatibility makes a commitment that a weighted-penalty model doesn't: no amount of entrenchment can make the same clashing operator assignment grammatical. Apparent change in cases like **I've finished it yesterday* should come either from read-out effects or from re-licensing. Where present-perfect combinations with definite past-time adverbials are attested in a community (Werner, 2013a, 2013b), change has to proceed by *re-licensing*: a perfect construction with a different operator contribution enters the repertoire, potentially producing judgment distributions that are bimodal across communities and categorical within speakers, not gradual within-speaker reduction in the status cost of the original clash. A soft-constraint model predicts the opposite signature. Judgment-distribution data from the relevant contact varieties decides it.

This contrasts with preempted gaps. A preempted gap can re-open when changed preferences, opportunities, or target evidence move the population across a transition boundary: average licensing then rises through the middle, producing within-speaker gradient during the transition and potentially a community S-curve.

Because production supplies later evidence and retained evidence affects later adoption, some settings permit a stronger, history-dependent pattern. Two communities under the same current

external conditions can remain stably different because their inherited evidence states place them on different trajectories. Licensing can then disappear at one point as conditions worsen but return only after conditions improve beyond a different point. This history dependence predicts distinct loss and recovery boundaries.

The prediction is conditional and requires longitudinal evidence. Effective exposure and the evidence speakers require before adopting or retaining the form have to be estimated without using the transition points; if matched loss and recovery trajectories retrace the same boundary, a steep but reversible response is the better account.

Compatibility failures can't re-open that way; they require a distinct construction with a different operator contribution. The identification discipline is the same as elsewhere: the re-licensed construction has to be diagnosed before the judgment data is used, for example by showing that it supports other definite past-time adverbials, scope interactions, or ellipsis patterns predicted by the new operator contribution.

INTONATIONAL OPERATORS. Nothing in the model privileges segmental material. Intonational patterns are operator candidates where independent evidence establishes their paradigms and assigns them a high public-update profile. English polar-question rises and nuclear-accent placement are candidate cases, not classifications supplied by their familiarity (Reynolds, 2026b). The Turkish analysis in Supplementary File 2 instead isolates segmental exponent licensing: it doesn't establish that the containing tense or number paradigm is an operator. After opportunity and competitor strength are matched, mis-set intonational operator values should produce more update confusion than value-intact exponent errors or non-operator accent shifts. Categorical correction and resistance to framing can occur in all three cases, so neither response diagnoses an operator profile by itself.

6.2 MACRO-TYPOLOGICAL CONSTRAINTS ON GRAMMATICAL DESIGN

Macro-typological work shows both recurrent constraints and substantial community-specific freedom. Verkerk et al. (2025) test 191 implicational universals against Grambank's 2,430-language sample while controlling for genealogical and areal non-independence. Only about a third remain statistically supported; the survivors concentrate in hierarchical and relatively narrow word-order patterns, often in configurations that languages are more likely to evolve into than out of.

OVMG treats such results as candidate attractors in the design space, not as proof that a comparative feature is a language-internal grammatical object. That stronger claim requires a comparandum-to-realization mapping: the feature, its realization in each language, and the held-out language or diachronic outcome to be predicted have to be specified (Reynolds, 2025). Recurrent configurations may reflect shared cognitive and communicative pressures, while the many unsupported universals leave room for contingent community conventions. Comparative evidence locates candidate stable regions; OVMG supplies conditional routes by which local licensing could reach them.

6.3 RELATIONSHIP TO GENERATIVE GRAMMAR

Generative work established that grammaticality can't be reduced to semantic plausibility, processing ease, or attestation frequency. *Colorless green ideas sleep furiously* is structurally well formed despite its bizarre meaning, while English left-branch extraction (**Which did you buy car?*) remains unacceptable despite its clear intent. OVMG retains this systematicity (Dennett, 1991) while distinguishing coverage, hard compatibility, saturation, and concentrated exclusion as possible sources.

A generative analysis supplies warrant when it specifies a contrast’s target, diagnostics, evidence, and failure conditions. OVMG targets the community licensing profile into which such structural contrasts enter. Thus the syntax of independent relative *whose* doesn’t decide whether its rare niche is conventionally licensed, just as structural possibility doesn’t decide which age-predication frame a community establishes (§5).

6.4 RELATIONSHIP TO CONSTRUCTION GRAMMAR

Construction Grammar (CxG) treats learned form–meaning pairings at every grain as grammatical objects (Fillmore, 1988; Fillmore et al., 1988; Goldberg, 1995, 2019; Kay & Fillmore, 1999; Sag, 2012); OVMG inherits that ontology and rejection of a clean core/periphery split. It also asks whether a construction is licensed, realizes a value in an identified paradigm, selects the intended value, and uses a licensed exponent. The paradigm’s public-update contribution is profiled separately. Licensing remains broader than the operator stratum: a community can exclude age-predication *have* or a head-specific complement without an operator error, while material at any structural level can realize a high-profile paradigm (Reynolds, 2026b). Opportunity predicts licensing evidence; public-update divergence predicts the narrower comprehension and repair profile.

6.5 RELATIONSHIP TO USAGE-BASED APPROACHES

OVMG inherits the usage-based premise that linguistic knowledge develops from patterns of use (Bybee, 2006, 2007, 2010). Its speaker/population split restates Schmid’s (2020) distinction between individual ENTRENCHMENT and community CONVENTIONALIZATION; its change process draws on utterance-selection theory (Blythe & Croft, 2012; Croft, 2000). Frequency per opportunity likewise joins Stefanowitsch’s (2006) treatment of significant absence with Labov’s principle of accountability (Labov, 1972).

OVMG integrates these resources and adds a prediction. It separates a candidate’s availability from a speaker’s choice among available candidates. That division puts a quantity on the absence that Stefanowitsch left open: he showed how to establish that a form is significantly absent but stressed that significant absence “does not, in itself, provide any clues as to why”. A significantly absent form may be unlicensed, licensed but dispreferred, or avoided together with its competitors, and those diagnoses project differently. The formal supplement states the factorization and the restricted conditions under which an attributable omission provides approximately direct evidence of non-licensing.

Speaker-level evidence estimates feed the adoption–retention process described in §4.3. The relation to Goldberg’s (2019) COVERAGE and statistical preemption is one of formalization in the same spirit. Goldberg’s “coverage” is overall frequency support; structural coverage here asks whether a complete analysis exists. Her “explain me this” puzzle (a high-coverage double-object frame preempted by the prepositional dative) is a case in which accumulated, attributable competitor wins exclude the target in a high-opportunity niche.

The dative alternation supplies the both-licensed control. Bresnan et al. (2007) model choice between *give Kim the book* and *give the book to Kim*; a frozen model could be rescored on Spoken BNC2014 conversation. Its probability would estimate recipient realization conditional on a token already inside the annotated alternation, not grammaticality. Reaching the broader licensing set requires a different denominator and evidence stream.

The proposed rescore would test whether length, animacy, definiteness, and pronominality transport while verb base rates vary, using both discrimination and calibration (Goodman, 1955). It hasn’t

been run. More tokens from the same annotated sample still couldn't distinguish an unlicensed gap from a dispreferred licensed form.

One caution follows from Ambridge et al. (2015), who find that overall form frequency (entrenchment) predicts the retreat from overgeneralization more robustly than competitor frequency (preemption), and who propose collapsing the two into a single competition process. OVMG keeps them analytically separate as direct error observations and counterfactual choice observations. They have to be analysed jointly rather than added as interchangeable failure counts (§4.1). Whether one underlying competition process accounts for both remains an empirical question rather than a stipulation.

Independent relative *whose* supplies the complementary low-opportunity case (§5). OVMG thus keeps the usage-based role for frequency and entrenchment while predicting when rarity is informative preemption and when it remains uncertainty.

6.6 RELATIONSHIP TO LOGICALITY OF LANGUAGE ACCOUNTS

Logicity-of-language accounts explain some restrictions by positing a system that identifies meanings made trivial by logical or closed-class vocabulary (Chierchia, 2013; Fox, 2000; Gajewski, 2009). Their central challenge is that ordinary tautologies and contradictions can be acceptable. To address this, Del Pinal (2019) allows context-sensitive rescaling of open-class terms: the second occurrence of *raining* in *It's raining and it isn't raining*, for example, can be understood as 'raining hard'.

This account targets restrictions tied to logical vocabulary. OVMG also has to distinguish absent analyses (1a), conventional exclusions (5), and rare, weakly evidenced constructions (1d). It treats interpretive rescaling as one possible route to a compatible analysis, while keeping that analysis separate from community licensing and subjective processing effects (§3.6). The two approaches overlap on some compatibility cases but make different commitments about the source and scope of grammatical exclusion.

6.7 RELATIONSHIP TO RELEVANCE-THEORETIC ACCOUNTS

Scott-Phillips (2026) argues that acceptability intuitions arise as by-products of systems for interpreting communicative acts, with unacceptability reflecting an inability to interpret an utterance consistently with presumptions of communicative efficiency. OVMG shares the view that communicative practice shapes linguistic conventions, but treats the resulting form-value relations as constraints that needn't reduce to present efficiency.

The empirical difference concerns what has to be established independently. A relevance-theoretic account has to specify why a case is inherently impossible to interpret rather than merely difficult. OVMG has to identify the relevant construction, situation, and any operator clash before using the judgment as evidence. Novel constructions becoming acceptable or unacceptable provide the clearest comparison: an efficiency account predicts movement with changing interpretability, whereas OVMG allows interpretability to remain high while community licensing or operator compatibility changes.

7 LIMITATIONS AND FUTURE DIRECTIONS

Five main limits bound the account. First, a "communicative situation" combines immediate setting, social position, and an oriented-to norm centre; operationalizing those overlapping dimensions requires more precise methods.

Second, the formal model uses an approximate evidence filter, declares rather than estimates adoption, and approximates the expected population transition. Its bounded-memory results remain unproven; networks, cohort replacement, and a shared observation model remain future work. Emergence is conditional on these interfaces, not a generic consequence of learning.

Third, most stylistic choices aren't grammaticality facts. The account has to explain when situation-linked preferences begin to affect licensing, repair, or categorical rejection.

Fourth, restrictions that resist compatibility, saturation, and preemption analyses test whether a complete covering analysis exists or whether the constructional inventory was drawn too narrowly.

Finally, the model asks speakers to retain situation-indexed information about forms, values, neighbours, opportunities, competitors, and repairs. Compositional inheritance and pooling may compress that burden, but neither process has been estimated.

7.1 METHODOLOGICAL CONSIDERATIONS

Experimental syntax supplies tools for separating the proposed evidence streams (Schütze, 2016). Corpus rates have to be normalized by independently specified opportunities; judgments should be accompanied by confidence, test–retest, and exposure measures; and processing measures can test whether practice lowers implementation cost while production and repair remain stable. Sociolinguistic fieldwork fixes the relevant community and norm centre, while longitudinal and historical data test whether evidence concentration changes before categorical treatment does. No one channel identifies licensing alone.

A DECISION-LAYER DISCONTINUITY CHECK. A sudden change from acceptance to repair, clarification, or sanction near an independently fixed practical boundary would support a response switch rather than a discontinuity in licensing itself. A smooth response would weaken that switch, not the licensing-state theory. Supplementary File 1 gives the confirmatory design and its failure condition.

8 CONCLUSION

Grammaticality, de-idealized, isn't one fact about an ideal speaker. It's a conditioned profile: whether some analysis covers a form and its intended value, whether its operator contributions unify, whether its obligatory contrasts are supplied, and whether the relevant population licenses its constructions. A speaker's evidence about that profile and the resulting feelings of anomaly and confidence are separate objects. So are genuine differences among speakers and mere uncertainty about the population mean.

The contribution is the integrated architecture and the dissociations it makes testable, not the claim that each component is unprecedented. Weakly evidenced forms should respond differently to framing and exposure from forms repeatedly defeated in genuine opportunities; licensed but processing-heavy forms should improve without a change in licensing; moribund contrasts should lose evidential support before their categorical treatment disappears; and wrong-value errors in independently identified, high-profile paradigms should draw a different repair profile from value-intact constructional errors. These claims are informative only if situations, niches, alternatives, and target projections are fixed before the outcomes are inspected.

Within the model, and only under the stated adoption and retention conditions, the population dynamics establish the possibility of separated licensing regimes. Evidence updating changes estimates but can't change population composition unless coupled to adoption or retention. Repair is a candidate maintainer or controller, not one by definition; maintenance requires transmission evidence, and control requires a deviation-sensitive intervention that restores the threatened relation. A history-dependent loss-and-recovery pattern is likewise predicted only where two stable regimes and their external controls have first been established.

The resulting account is less simple than a single grammaticality score, but its recurring distinctions handle cases that otherwise demand unrelated explanations. Their economy has to be earned empirically: estimate licensing from converging non-rating indicators, normalize production by independently specified opportunities, and let the projections fail when their indicators are wrong. De-idealizing grammaticality thereby replaces one idealized variable with a profile that answers to evidence and predicts which rejections soften, which forms transmit, and which absences hold.

DATA AVAILABILITY/SUPPLEMENTARY FILES

Supplementary File 1 gives the mathematical specification, derivations, simulation contract, decision-layer design, and machine-checked structural correspondence; the machine check doesn't cover dynamics or empirical identification. Supplementary File 2 develops the Turkish vowel-harmony case, including how licensing confined to narrow borrowing situations can become ordinary lexical licensing. Supplementary File 3 is a reproducibility archive containing simulation code, declared parameters and seeds, aggregate outputs, formalization sources, and replay instructions. Raw COCA concordance lines aren't redistributed; access remains subject to the English-Corpora terms. The archive instead provides the registered agreement queries, coding protocol, and aggregate counts. The exploratory independent-relative *whose* search is documented as an unregistered zero-result illustration whose exact interface query wasn't preserved and which supplies no licensing estimate.

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COMPETING INTERESTS

The author has no competing interests to declare.

AI DECLARATION

Anthropic Claude (including Fable and Opus), OpenAI ChatGPT/Codex, Google Gemini, DeepSeek, and Ox Alpha through OpenCode Zen were used during 2025–2026 for brainstorming, drafting and editing assistance, adversarial review, literature triage, and code or formalization review. Aristotle assisted with Lean formalization. The author selected and revised all prose and claims and checked outputs bearing on claims, as applicable, through primary-source reading, recomputation, executable tests, or local Lean replay. Model reviews identified targets for those checks; they weren't treated as evidence. These systems supplied no participant data and didn't independently establish empirical results. The author takes responsibility for the submitted work.

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